


**Institute of Technology of Cambodia**

**Department of Information and Communication Engineering**




# Internet Programming I

## Group: I4-GICA

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# Project Proposal

## 1. Project Overview

### 1.1 Project Title (ScholarTrack)

ScholarTrack is a web-based application designed to help students discover, organize, and manage scholarship opportunities and academic competitions in one centralized platform. The system aims to simplify the process of finding relevant opportunities, keeping track of application requirements, and monitoring important deadlines. Instead of relying on multiple websites, social media platforms, or personal notes to manage information, students can use ScholarTrack to access opportunity details and organize their application activities more efficiently.

### 1.2 Introduction

Scholarships and academic competitions provide students with opportunities to continue their education, develop their skills, and achieve their academic and professional goals. However, information about these opportunities is often distributed across different websites, social media platforms, and institutional announcements. As a result, students may find it difficult to organize opportunity information, remember application deadlines, and monitor the progress of multiple applications.

The system also provides an administrator interface for managing opportunity information and ensuring that published listings are organized and up to date. ScholarTrack will be developed as an individual project using Vue 3 with TypeScript for the frontend, NestJS for the backend, PostgreSQL for database management, and Prisma as the object-relational mapping (ORM) tool.

### 1.3 Problem Statement

Students often receive information about scholarships and competitions from multiple sources, making it difficult to manage opportunities efficiently. Important information, such as eligibility requirements, application deadlines, and required documents, may be scattered across different platforms.

Furthermore, students may rely on personal notes or spreadsheets to track their applications. These methods can become difficult to maintain when managing several opportunities simultaneously, increasing the risk of forgetting deadlines or losing track of application progress.

Therefore, a centralized web-based system is needed to help students organize opportunity information, monitor deadlines, and manage their application progress in a structured and convenient manner.

### 1.4 Project Objectives

The main objective of ScholarTrack is to develop a web-based platform that simplifies the discovery and management of scholarship and competition opportunities.
The specific objectives are:

- To provide a centralized catalogue of scholarship and competition opportunities.
- To enable students to search and filter opportunities based on keywords, categories, and deadlines.

- ❖ To allow students to save opportunities for future references.
- ❖ To provide application tracking features that allow students to manage their application statuses.
- ❖ To display upcoming deadlines and application statistics through a personalized dashboard.
- ❖ To provide administrators with tools to create, update, publish, and manage opportunity listings.
- ❖ To implement secure authentication, role-based authorization, input validation, and data persistence.
- ❖ To develop a maintainable and testable application using the technologies required by the course.

### 1.5 Project Scope

The project focuses on developing the core features required for students to discover opportunities and manage their applications.
In scope:

- ❖ User registration and login.
- ❖ Student and administrator roles.
- ❖ Opportunity listing and detailed information pages.
- ❖ Opportunity search, filtering, and pagination.
- ❖ Saving and removing saved opportunities.
- ❖ Creating, updating, and deleting personal application records.
- ❖ Application status management.
- ❖ Dashboard statistics and upcoming deadlines.
- ❖ Administrator management of opportunity listings.
- ❖ Database integration, input validation, authorization, and automated testing.

## 2. System Requirements Specification

### 2.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements of ScholarTrack. It describes the system's expected behavior, user roles, interfaces, data requirements, security constraints, and testing criteria. The document will serve as a reference during system design, implementation, testing, and evaluation.

### 2.2 Intended Audience

This document is intended for:

- ❖ The project developer, who will design, implement, and test the application.
- ❖ The course instructor, who will review and evaluate the proposed system.
- ❖ Students who will use the application to discover opportunities and track applications.
- ❖ Administrators who will manage the opportunity catalogue.

### 2.3 Product Perspective

ScholarTrack is a client-server web application consisting of three main layers:

1. Frontend: Provides the user interface and handles user interactions.
2. Backend: Processes requests, enforces business rules, validates input, and manages authentication and authorization.
3. Database: Stores user accounts, opportunity information, saved opportunities, and application records.

## 2.4 User Classes and Characteristics

**Student**
Students are the primary users of ScholarTrack. They can:
- Register and log in to the system.
- Browse and search published opportunities.
- View opportunity details and eligibility information.
- Save opportunities for later reference.
- Create and manage their own application records.
- Update application statuses and personal notes.
- View application statistics and upcoming deadlines.

**Administrator**
Administrators are responsible for maintaining the opportunity catalogue. They can:
- Log in to the administrative interface.
- Create new opportunity listings.
- Edit existing opportunity information.
- Publish or archive listings.
- Delete listings when appropriate.
- Maintain accurate deadlines, categories, and source links.

# 3. Functional and Non-Functional

- **Functional**

| Functional Requirement | Description |
|---|---|
| User Authentication | The system allows users to register, log in, and log out securely. |
| Role Management | The system supports Student and Administrator roles with different permissions. |
| Opportunity Management | Students can browse, search, filter, and view opportunities. Administrators can create, update, and delete or archive listings. |
| Saved Opportunities | Students can save, view, and remove opportunities from their personal lists. |
| Application Tracking | Students can manage application records, update application statuses, and add personal notes. |
| Dashboard | The system displays application statistics, saved opportunities, and upcoming deadlines. |
| Data Validation | The system validates user inputs and displays clear error messages when information is invalid. |
| Data Security | The system protects user information and prevents unauthorized access to personal records. |

❖ **Non-Functional**

| Non-Functional Requirement | Description |
|---|---|
| Performance | The system should load pages and process user requests within a reasonable time under normal usage. |
| Security | The system shall protect passwords and user data and enforce role-based access control. |
| Usability | The system shall provide a simple, clear, and user-friendly interface that students can use easily. |
| Reliability | The system shall handle errors properly and prevent invalid operations from corrupting stored data. |
| Maintainability | The system shall use a modular structure and organized code to make future updates easier. |
| Compatibility | The system should work on modern web browsers, including Google Chrome, Microsoft Edge, and Firefox. |
| Data Integrity | The system shall maintain accurate and consistent data across related database records. |
| Responsiveness | The interface should adapt to different screen sizes, including desktops, tablets, and mobile devices. |

# 4. System Design and Architecture

## 4.1 System Architecture

ScholarTrack will use a three-layer architecture consisting of the presentation layer, application layer, and database layer. These layers work together to provide a maintainable and organized web application.

| Layer | Technology | Description |
|---|---|---|
| Presentation Layer | Vue 3, TypeScript | Provides the user interface for students and administrators. |
| Application Layer | NestJS | Handles business logic, authentication, validation, and API requests. |
| Database Layer | Docker | Stores and manages user accounts, opportunities, saved listings, and application records. |

### 4.2 Database Design

The system will use the following main entities to organize its data.

| Entity | Description |
|---|---|
| Users | Stores user information, login credentials, and roles. |
| Opportunities | Stores scholarship and competition details, including deadlines and eligibility requirements. |
| Saved Opportunities | Records the opportunities saved by each student. |
| Applications | Stores students' application statuses, notes, and related opportunity information. |

The system will support two types of users:
❖ Student: Can browse opportunities, save listings, track applications, and view the dashboard.
❖ Administrator: Can create, update, and manage scholarship and competition listings.

## 5. Conclusion

In conclusion, ScholarTrack: Scholarship and Competition Application Management System aims to provide a centralized platform for students to discover and manage scholarship opportunities and academic competitions. The system will help students organize important information, save opportunities, track application progress, and monitor deadlines more efficiently.

The project will implement essential features such as user authentication, role-based access control, opportunity management, application tracking, and a personalized dashboard. Using Vue 3, TypeScript, NestJS and Docker the system will be developed following appropriate software engineering principles.

Through this project, the developer will gain practical experience in full-stack web development, database management, system design, and software testing. Ultimately, ScholarTrack is expected to provide a user-friendly and reliable solution that helps students manage academic opportunities more effectively while demonstrating the practical application of software engineering concepts.